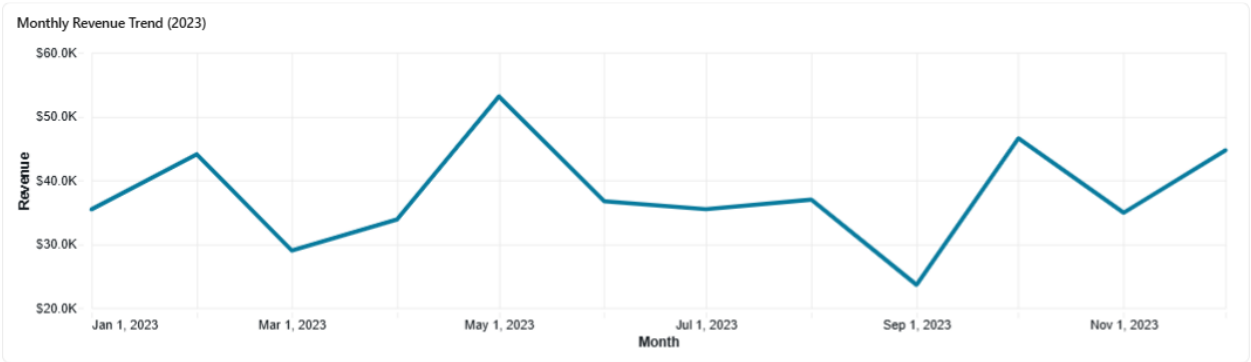


Time-Based Sales Analysis

The business generated \$456,000 in revenue across 1,000 transactions from January 2023 to January 2024, with May 2023 as the peak month (\$53,150) and September 2023 as the lowest (\$23,620).

Overall Temporal Patterns

The dataset spans 345 days from January 1, 2023 to January 1, 2024, with an average of 2.9 transactions per day. Monthly revenue shows significant volatility, ranging from \$23,620 to \$53,150, with an average transaction value of \$456 across the year.



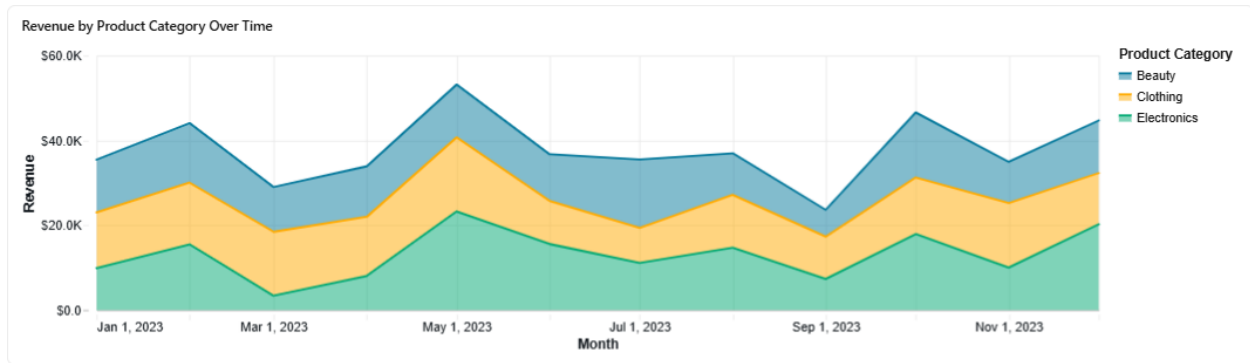
Monthly Revenue Trends

Peak Performance: May 2023 leads with \$53,150 in revenue (105 transactions), followed by October (\$46,580, 96 transactions) and December (\$44,690, 91 transactions). These months show strong year-end and mid-year performance.

Low Performance: September 2023 recorded the lowest revenue at \$23,620 (65 transactions), representing a 36% drop from August. March and April also showed weaker performance with revenues around \$29,000-\$34,000.

Volatility: The business experiences substantial month-to-month fluctuations, with revenue swings of up to \$29,530 between consecutive months (August to September).

Product Category Performance Over Time



Electronics shows the most dramatic swings, ranging from \$3,380 (March) to \$23,245 (May). The category dominates in May, December, and October, contributing significantly to peak months.

Clothing maintains more consistent performance throughout the year, leading in January, March, April, and November. Revenue ranges from \$8,250 (July) to \$17,455 (May).

Beauty peaks in July (\$16,090) and shows relatively stable mid-range performance, though it contributed to the September low with only \$6,320 in revenue.

Customer Behavior Patterns

Gender Trends: Revenue contribution by gender fluctuates significantly month-to-month. Female customers led in January, April, August, September, October, and December, while male customers dominated February, March, May, June, July, and November. Average transaction values vary widely, with male customers showing higher values in February (\$605) and female customers in October (\$619).

Age Group Patterns: The 35-44 and 45-54 age groups consistently generate the highest revenues in most months. The "Under 25" segment shows strong performance in February (\$13,435) and September (\$5,325), while the 55+ group peaks in May (\$14,330) and October (\$13,320).

Weekly Patterns



Saturday is the strongest sales day with \$78,815 in revenue (150 transactions) and the highest average transaction value of \$525.43. Monday and Tuesday also perform well with \$70,250 and \$69,440 respectively.

Thursday shows the weakest performance with only \$53,835 in revenue (123 transactions), followed by Sunday (\$58,600). The weekend effect is clear, with Saturday significantly outperforming weekdays.

Key Insights

1.

Seasonality exists but is irregular: Strong performance in May, October, and December suggests potential seasonal patterns, but the dramatic September dip breaks typical Q3 trends.

2.

Electronics drives volatility: The category's wide revenue swings (from \$3K to \$23K monthly) significantly impact overall business performance.

3.

Weekend advantage: Saturday's 34% revenue premium over Thursday indicates strong weekend shopping behavior, with higher transaction values suggesting more considered purchases.

4.

Diverse customer base: No single demographic dominates consistently, indicating broad market appeal but also potential opportunity for targeted campaigns during specific periods.